

Integrated Transcriptomic Analysis of Dilated and Ischemic Cardiomyopathy

Anas Jamshed

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Contents

Abstract	2
Introduction	2
Study Design	3
Target Genes	3
Materials and Methods	3
Data Acquisition	3
Gene Annotation	3
Dataset Integration	4
Metadata Harmonization	4
Batch Effect Correction	4
Differential Expression Analysis	4
Visualization	4
Differential Expression Results	4
Non-Ischemic Cardiomyopathy vs Control	4
Ischemic Cardiomyopathy vs Control	5
Principal Component Analysis	5
Non-Ischemic Cardiomyopathy	6
Ischemic Cardiomyopathy	7
Integrated Dataset	8
Heatmap Analysis	9
Control vs Non-Ischemic Cardiomyopathy	9
Control vs Ischemic Cardiomyopathy	10

Correlation Analysis	11
Volcano Plot Analysis	12
Non-Ischemic Cardiomyopathy	12
Ischemic Cardiomyopathy	13
Enhanced Volcano Plots	14
Ischemic Cardiomyopathy	14
Non-Ischemic Cardiomyopathy	15
Boxplot Analysis	16
Control vs Non-Ischemic CM	16
Control vs Ischemic CM	17
Biological Interpretation	17
Discussion	18
Conclusion	18
Reproducibility	18
Session Information	19

Abstract

Cardiomyopathy is a major cause of heart failure and cardiovascular mortality worldwide. This study integrates six independent Gene Expression Omnibus (GEO) microarray datasets to investigate transcriptomic alterations associated with ischemic and non-ischemic cardiomyopathy.

The workflow involved GEO data acquisition, probe annotation harmonization, Entrez Gene ID mapping, multi-study integration, batch effect correction using ComBat, differential expression analysis using Limma, and downstream visualization using PCA, heatmaps, correlation matrices, and volcano plots.

Introduction

Individual microarray studies often suffer from limited sample sizes and platform-specific biases. Integrating multiple independent datasets improves statistical power and biological robustness.

The following GEO datasets were analyzed:

GEO Series	Platform
GSE1145	GPL570
GSE1869	GPL96
GSE3585	GPL96

GEO Series	Platform
GSE3586	GPL3050
GSE9128	GPL96
GSE42955	GPL96

Study Design

The analysis compared:

- Control samples
- Non-Ischemic Cardiomyopathy (DCM)
- Ischemic Cardiomyopathy (ICM)

Target Genes

The study focused on several cardiomyopathy-associated genes.

Table 2: Cardiomyopathy Candidate Genes

Target_Gene
ACTN2
CRYAB
BMP10
CSRP3
DES
FHOD3
FLNC
LDB3
MYZAP
MYPN
MYOZ2
NEXN
PDLIM3
PDLIM5
TCAP
TTN

Materials and Methods

Data Acquisition

Public microarray datasets were downloaded from NCBI GEO using the GEOquery package.

Gene Annotation

Probe identifiers were converted into Entrez Gene IDs using AnnotationDbi and org.Hs.eg.db.

Dataset Integration

Expression matrices were aggregated at gene level and merged using common Entrez Gene IDs.

Metadata Harmonization

Clinical labels from all studies were standardized into:

- Control Group
- Non-Ischemic CM
- Ischemic CM

Batch Effect Correction

To minimize study-specific variation, ComBat from the sva package was applied.

Differential Expression Analysis

Differential expression analysis was performed using:

- Limma
- Voom normalization
- Empirical Bayes moderation

Visualization

The following visualizations were generated:

- PCA
- Heatmaps
- Correlation Heatmaps
- Volcano Plots
- Enhanced Volcano Plots
- Boxplots

Differential Expression Results

Non-Ischemic Cardiomyopathy vs Control

Table 3: Top Differentially Expressed Genes

GENE_SYMBOL	logFC	adj.P.Val
NFIB	-0.4601646	1.0e-07
VPS41	-0.5977663	1.0e-07
ZNF83	-1.3963356	1.0e-07
CETN3	-0.8947028	1.0e-07
TIA1	-0.4670973	1.0e-07

GENE_SYMBOL	logFC	adj.P.Val
PDE3B	-1.1752839	4.0e-07
PCOLCE2	-0.7779270	5.0e-07
HMGB2	-1.1023805	7.0e-07
PIGF	-0.5398995	7.0e-07
ACTR6	-0.8062493	7.0e-07
SNRPB	0.5483560	1.0e-06
PTP4A2	0.2721142	1.0e-06
AGPAT5	-1.3845535	2.7e-06
PLAGL1	-0.9554575	3.7e-06
MAPKAPK2	0.3127713	5.3e-06
CD59	0.2143394	5.6e-06
ATP7B	-0.7911946	7.2e-06
NISCH	-0.6545078	7.9e-06
EHHADH	-1.5214247	7.9e-06
PRKD1	-0.5179988	7.9e-06

Ischemic Cardiomyopathy vs Control

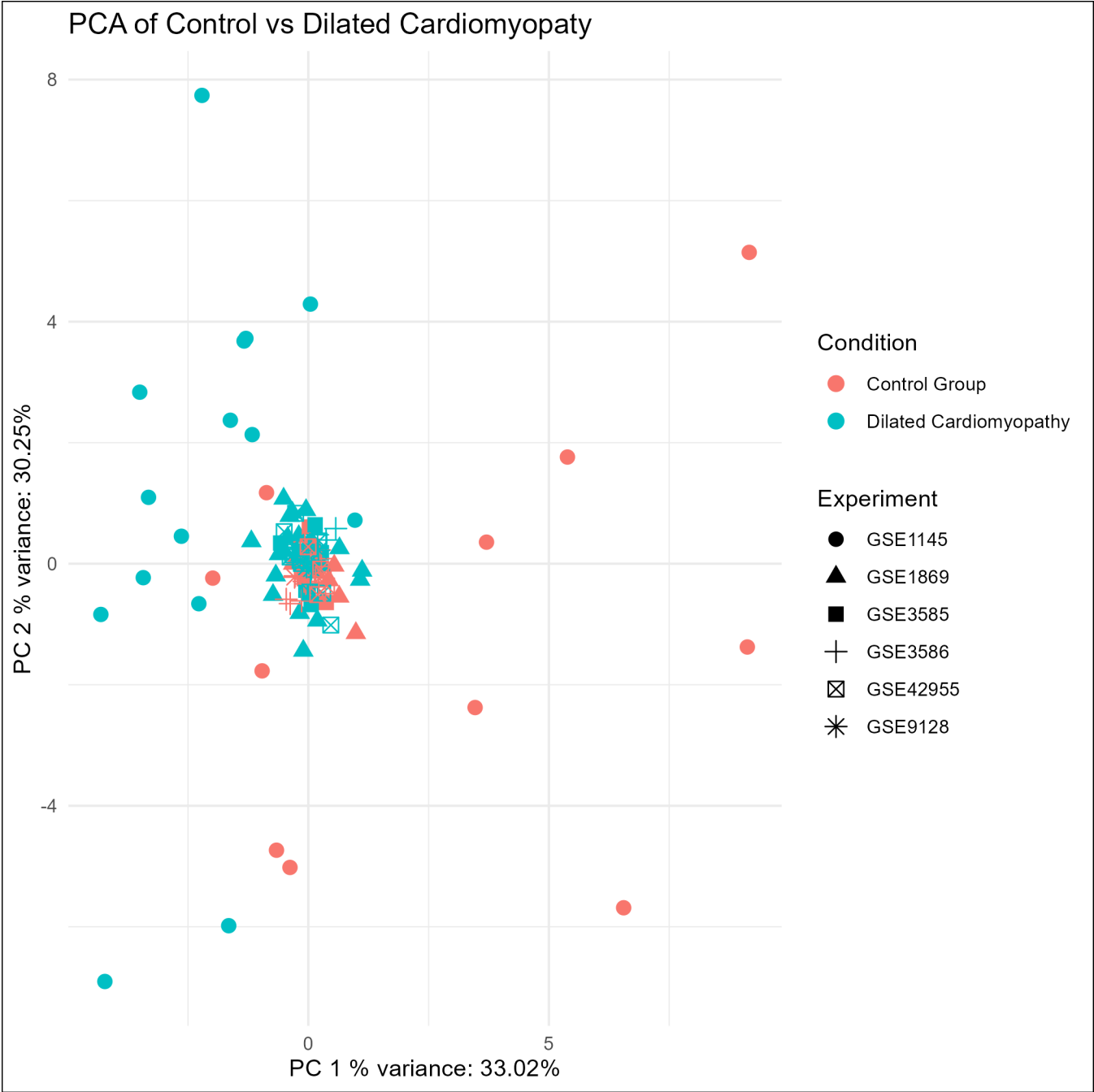
Table 4: Top Differentially Expressed Genes

GENE_SYMBOL	logFC	adj.P.Val
FCN3	1.8408550	0.0020201
SLC39A14	0.4723120	0.0066881
PLA2G2A	0.9921647	0.0066881
SH3BGRL3	0.8634155	0.0066881
PDE4D	0.7715972	0.0094094
SNRPB	0.4863341	0.0094094
ALDH9A1	-0.2497655	0.0098092
EGLN1	0.3728136	0.0100103
GTF2H5	-0.5014335	0.0100103
HSPB7	0.1948258	0.0100103
C1QTNF1	1.0481828	0.0100103
BRD3	-0.3207450	0.0137985
NCKAP1	0.6047742	0.0137985
CCL11	1.5860461	0.0137985
RANBP6	0.3331334	0.0137985
ZKSCAN1	-0.3663701	0.0137985
MSRB2	-0.2641410	0.0137985
ATP2A2	0.2184892	0.0157991
RPL37A	0.2484276	0.0157991
STRAP	0.2369115	0.0157991

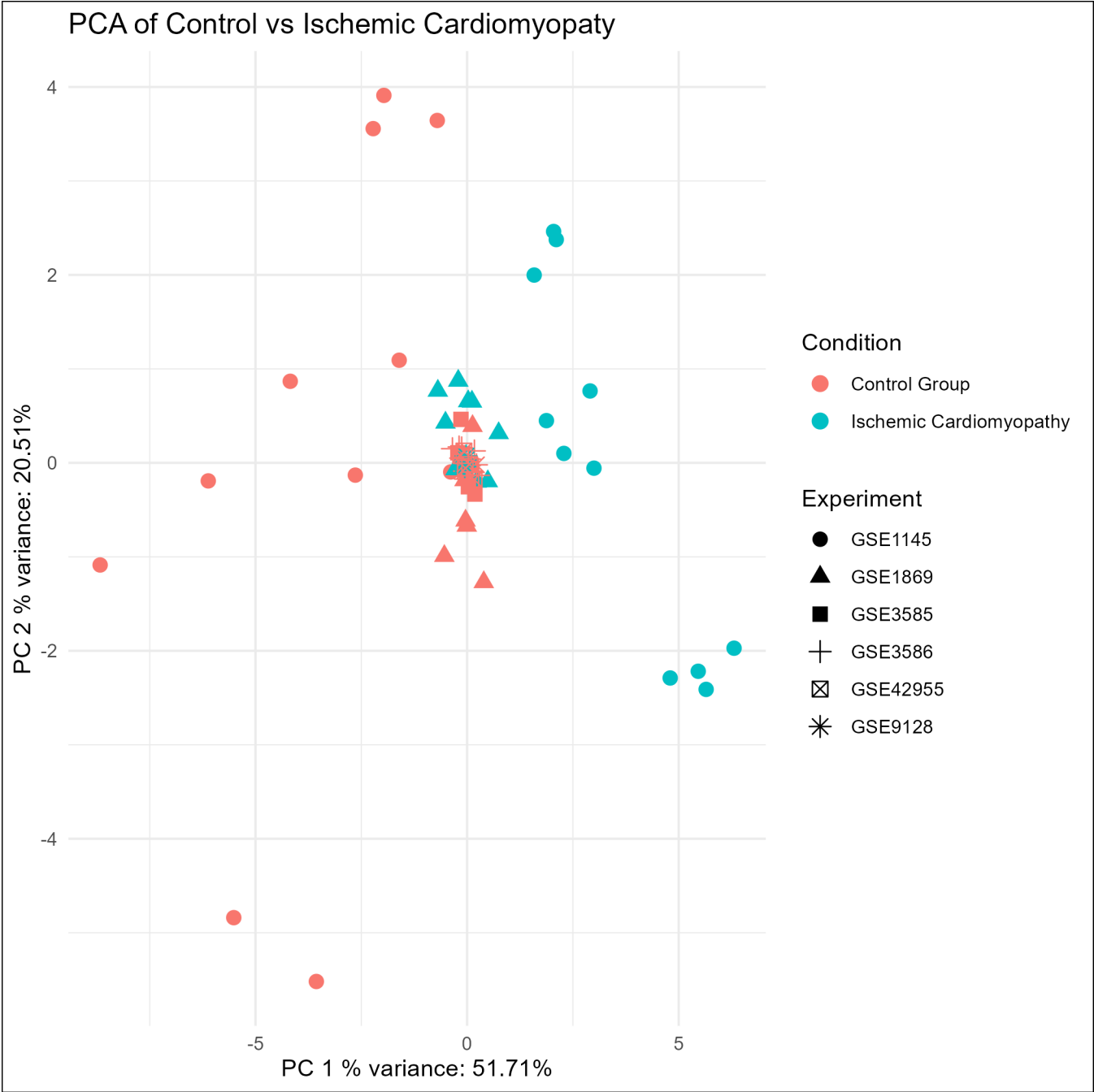
Principal Component Analysis

Principal Component Analysis was performed after batch correction.

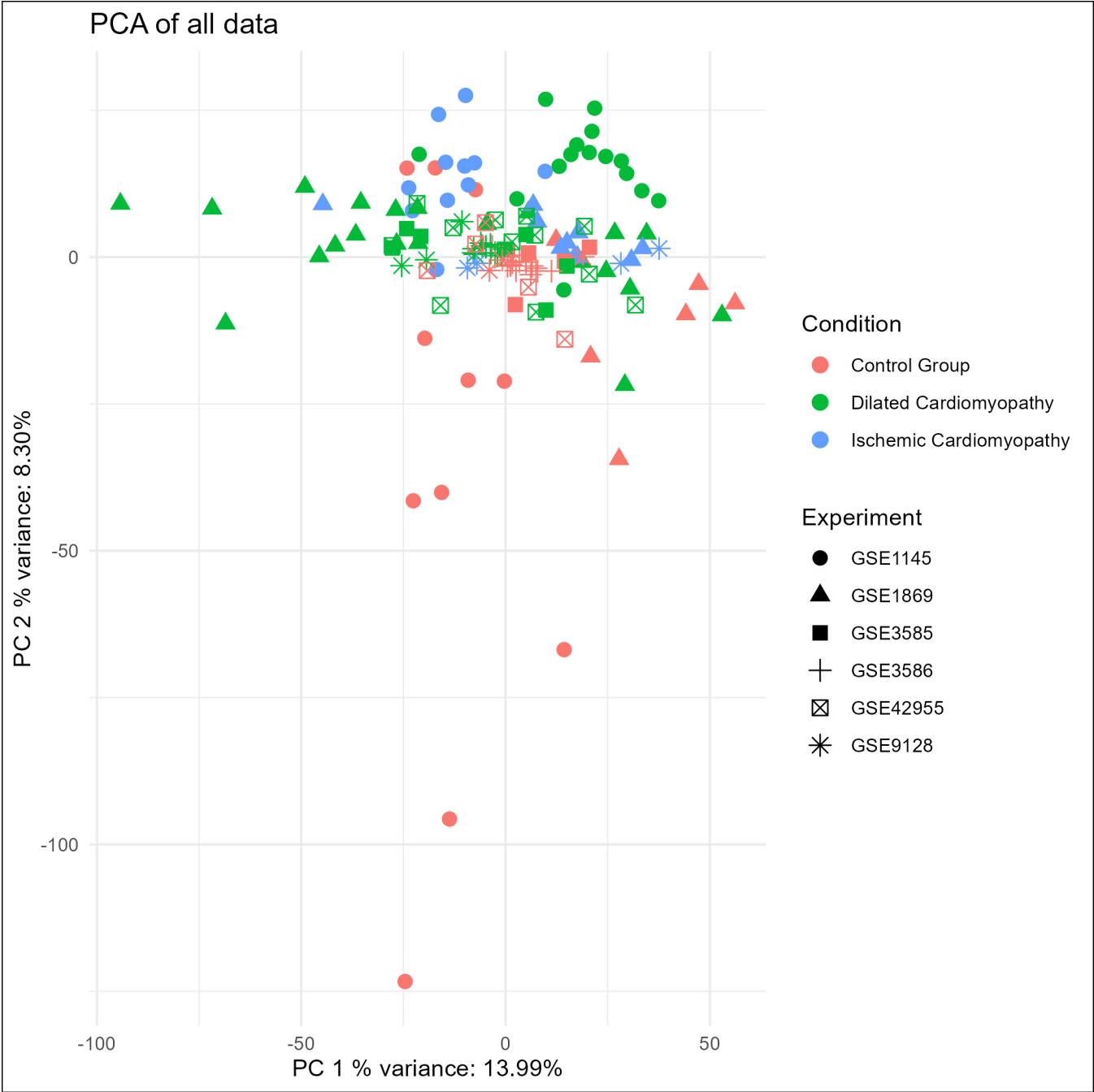
Non-Ischemic Cardiomyopathy



Ischemic Cardiomyopathy

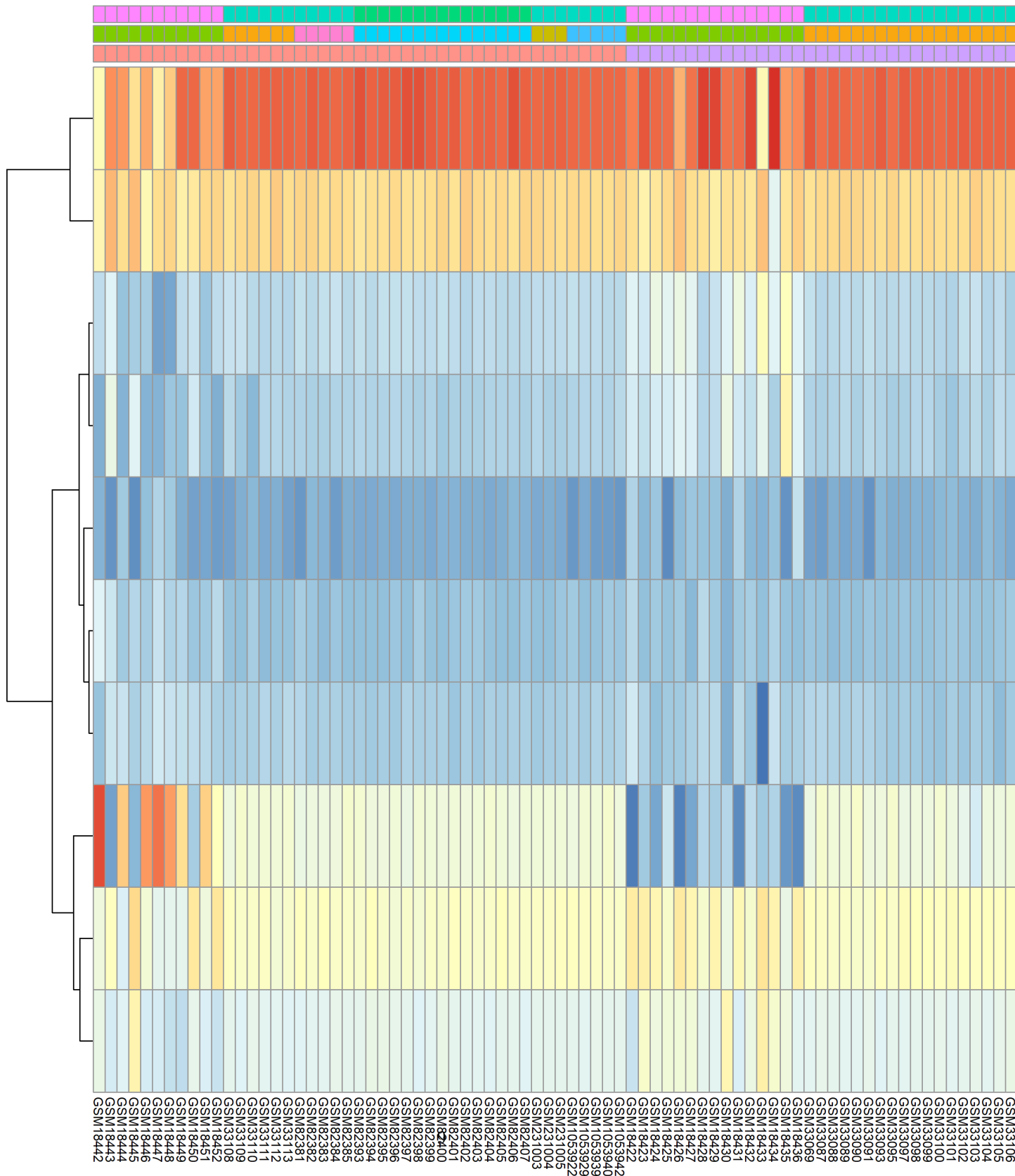


Integrated Dataset



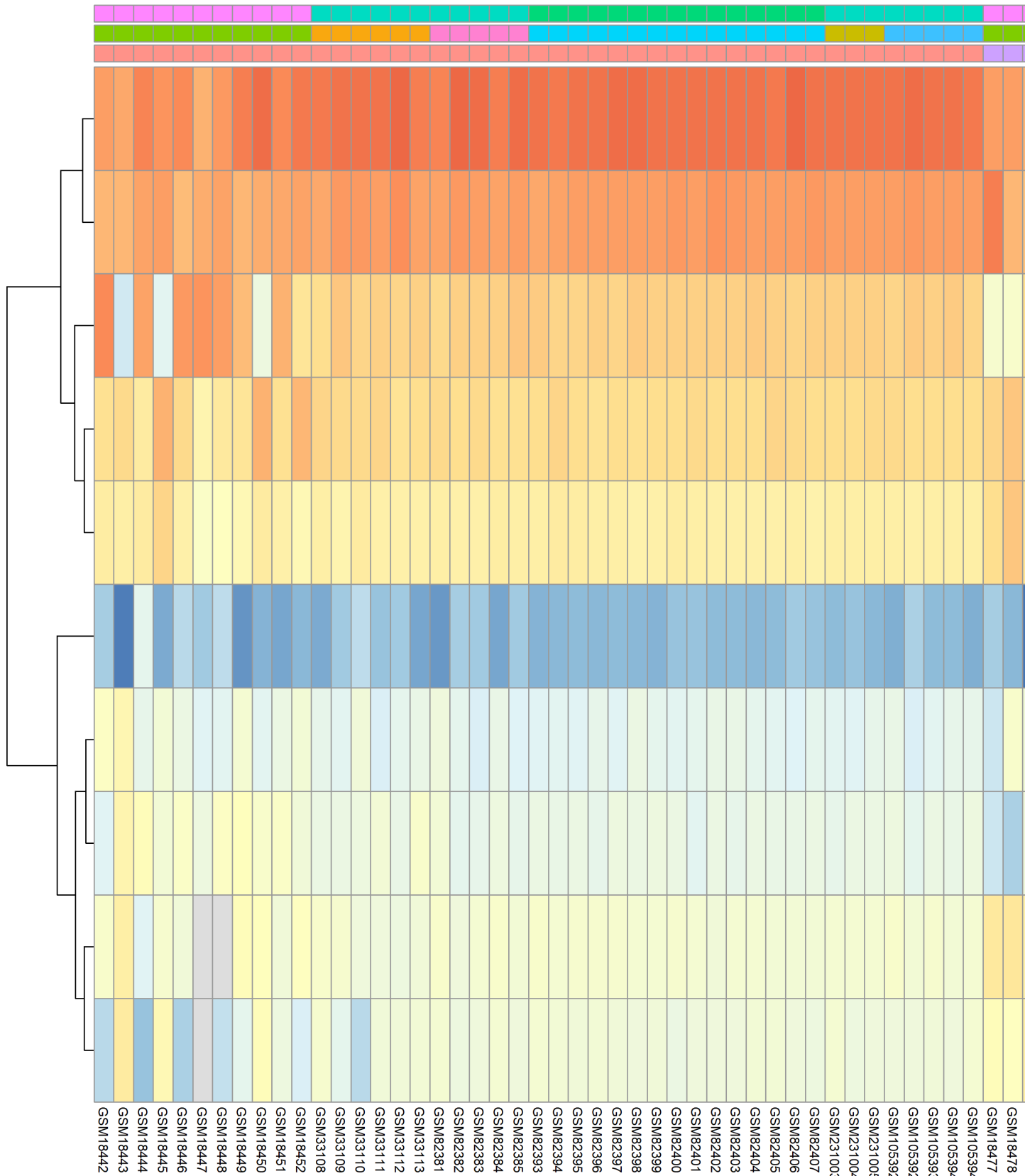
Control vs Non-Ischemic Cardiomyopathy

Heatmap of gene expression:

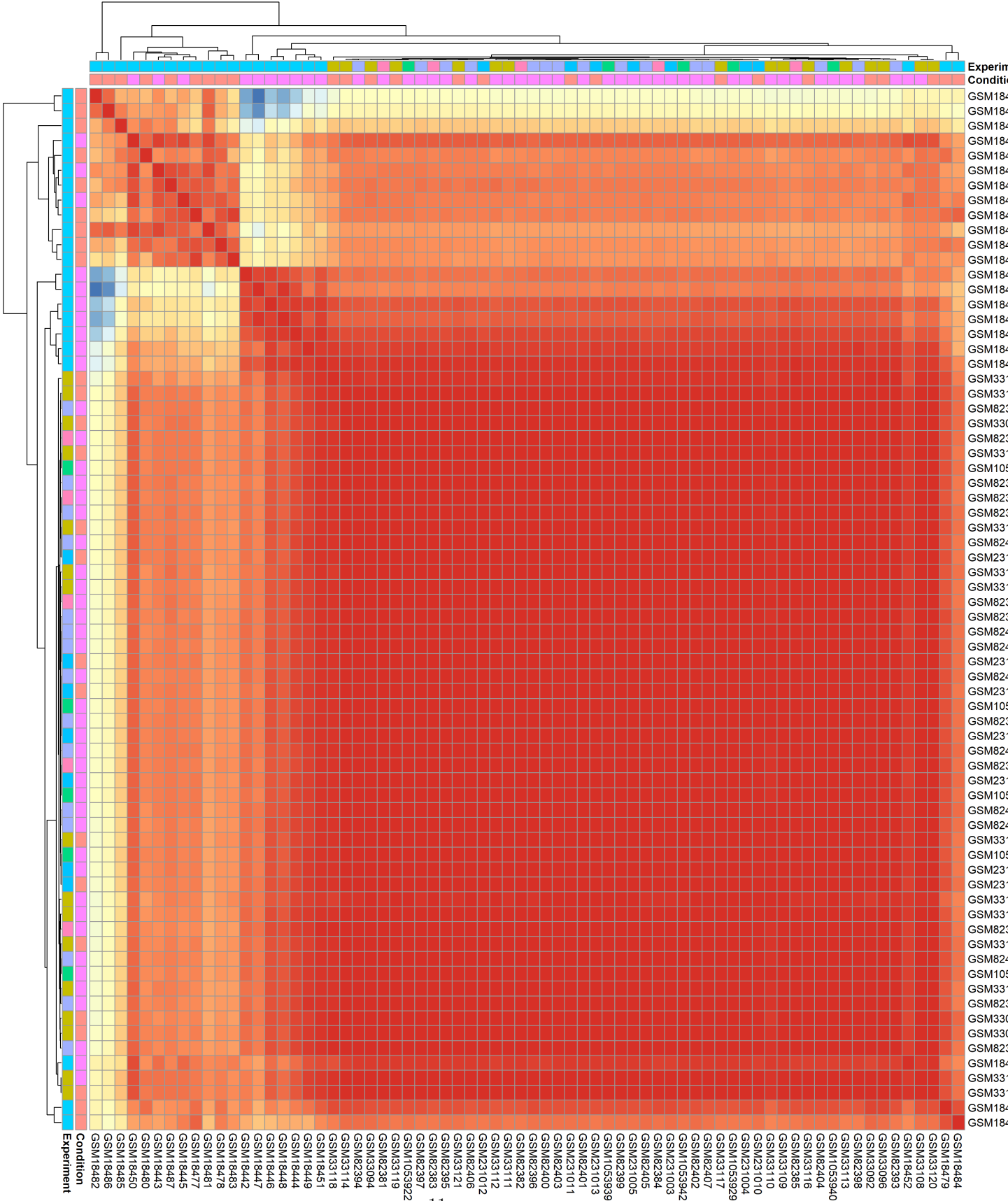


Control vs Ischemic Cardiomyopathy

Heatmap of gene expression:

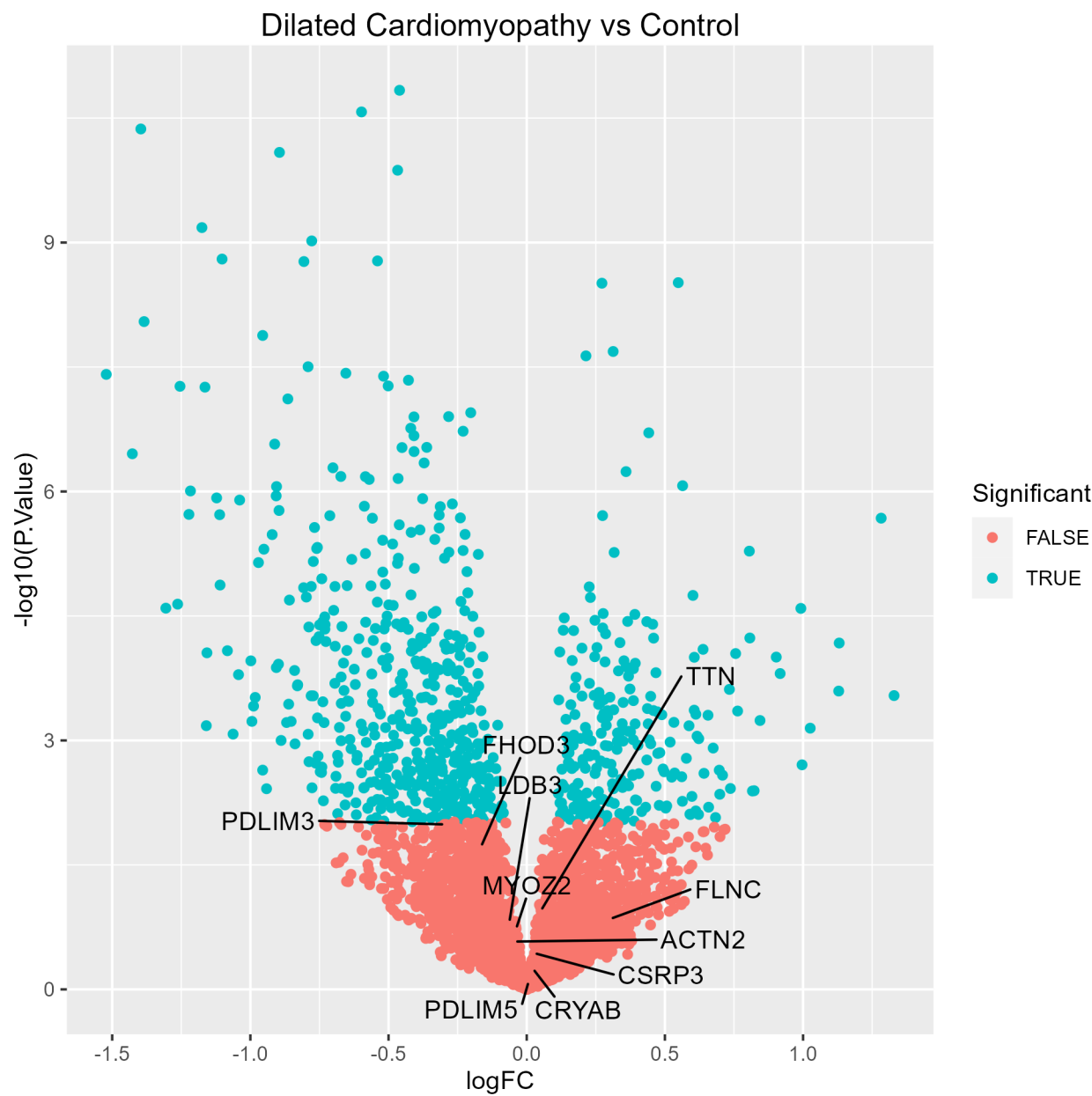


Correlation Analysis

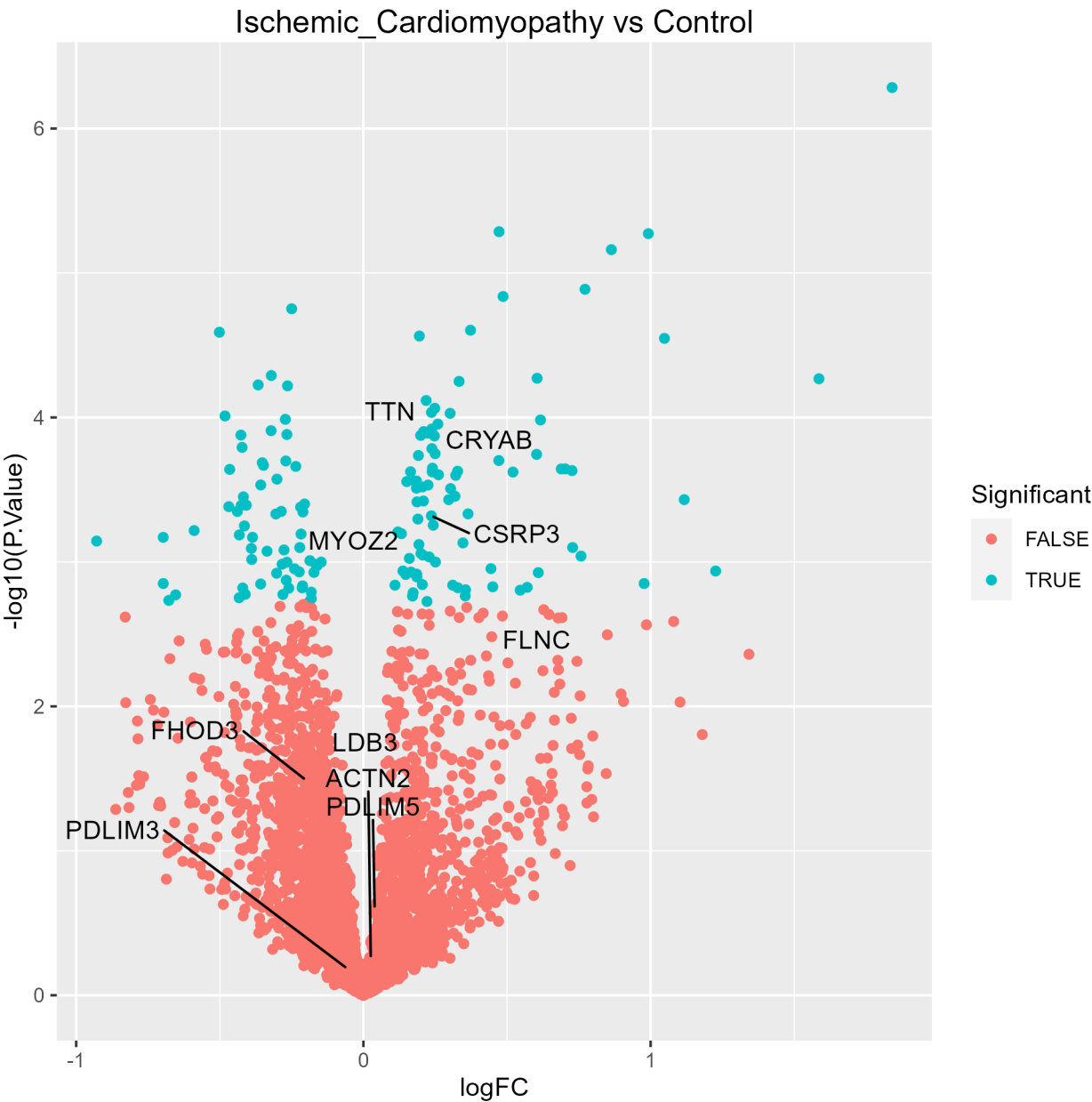


Volcano Plot Analysis

Non-Ischemic Cardiomyopathy



Ischemic Cardiomyopathy

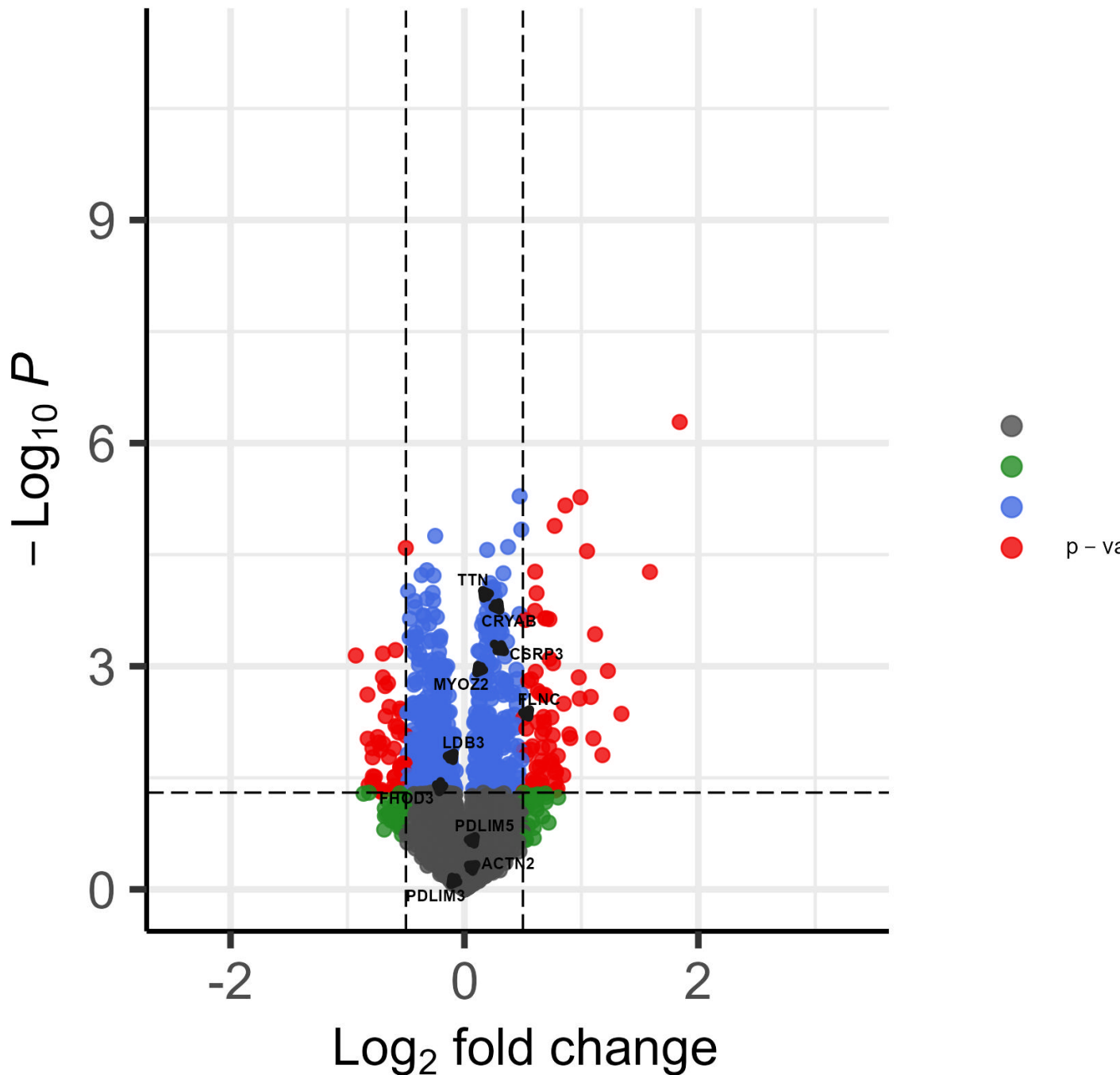


Enhanced Volcano Plots

Ischemic Cardiomyopathy

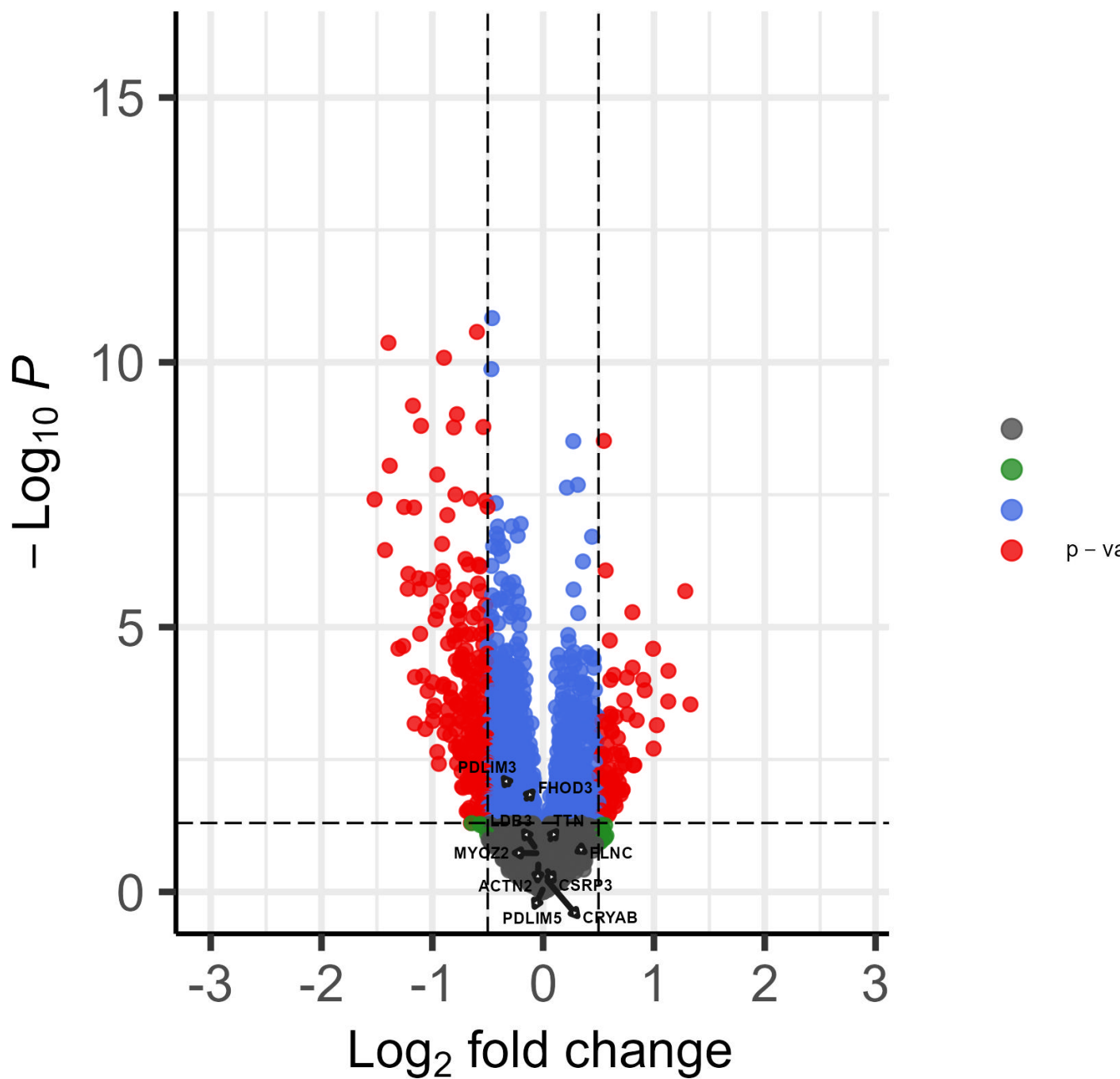
Ischemic Cardiomyopathy vs Control

Enhanced Volcano



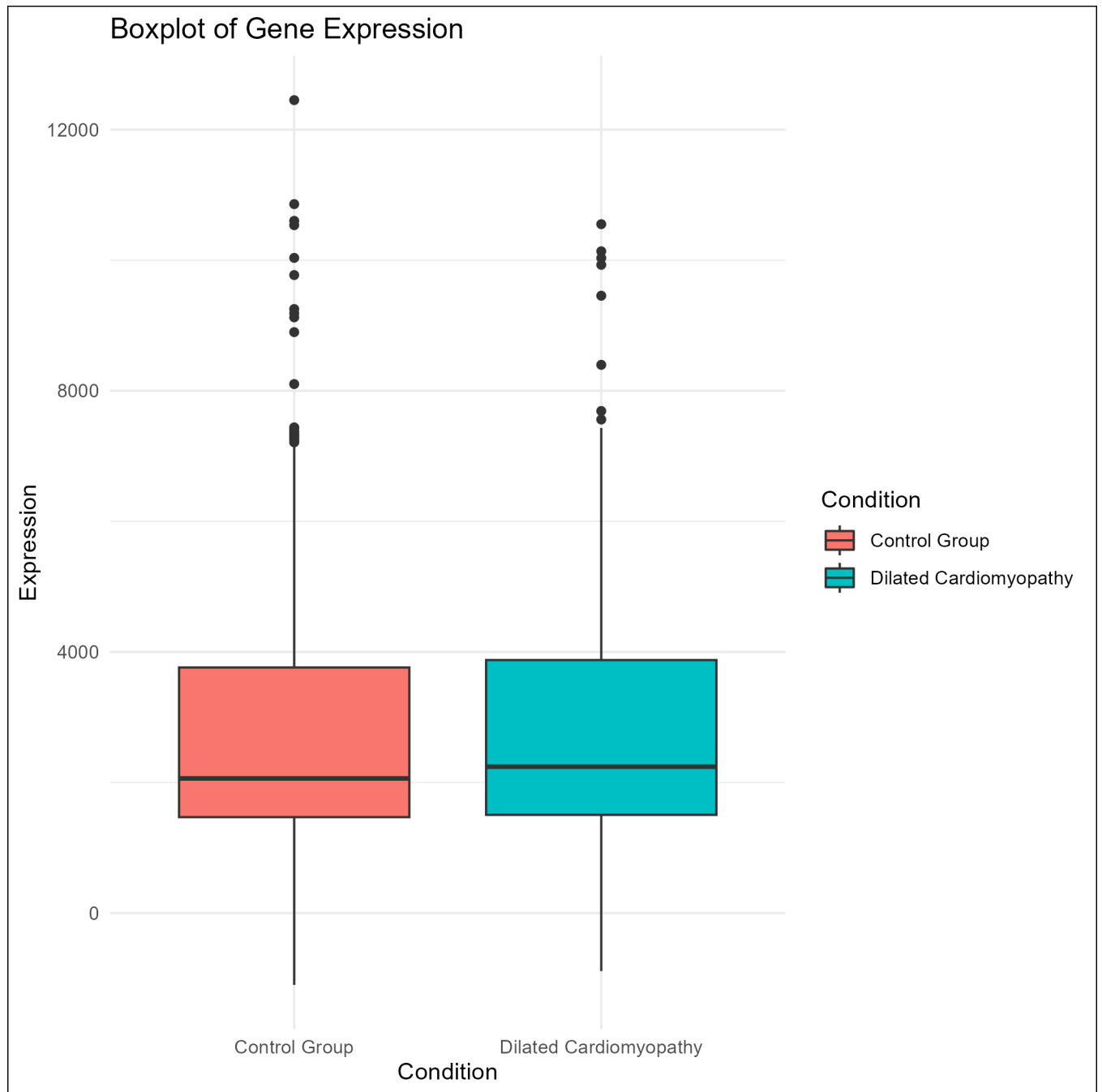
Dilated Cardiomyopathy vs Control

Enhanced Volcano

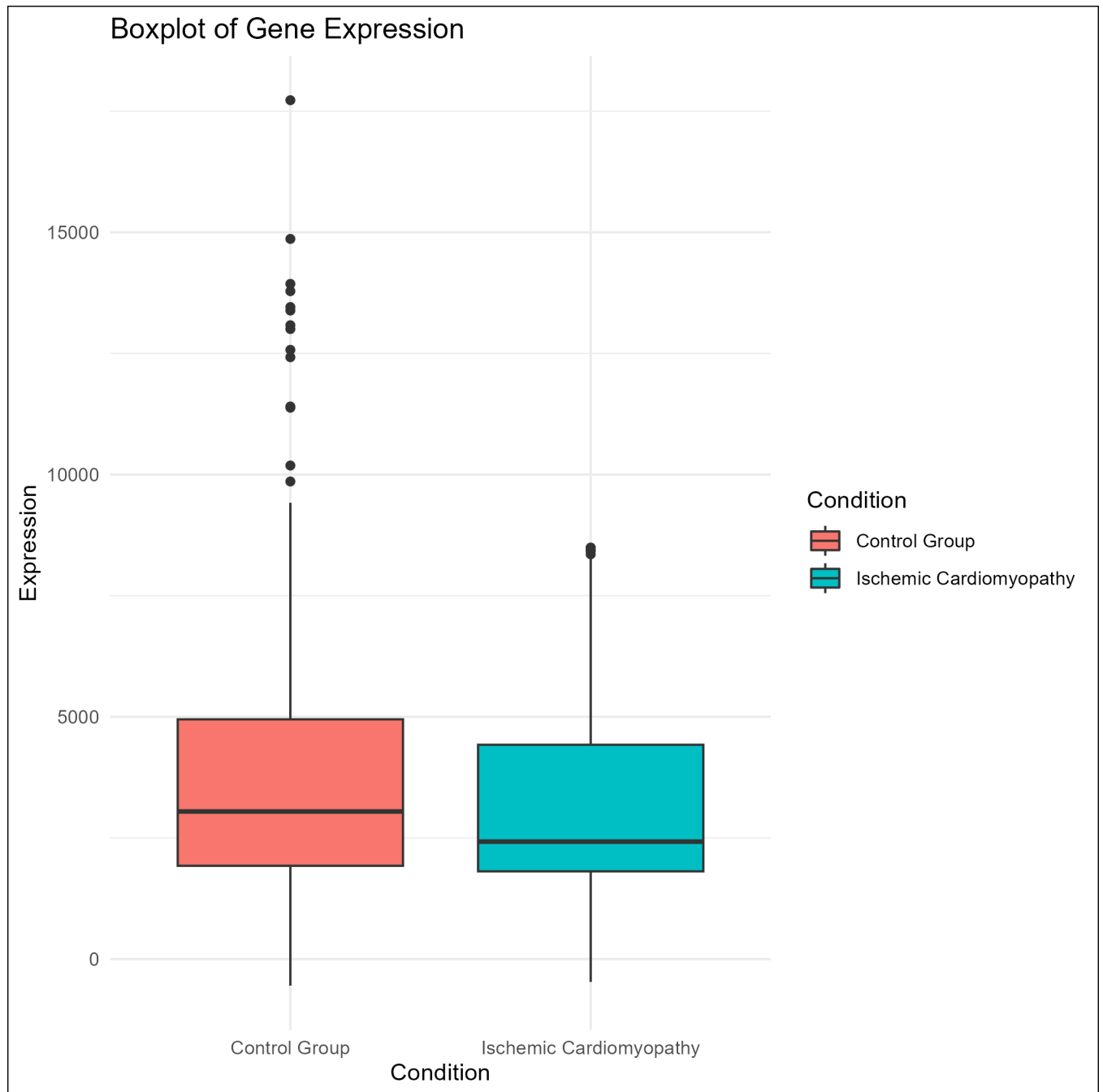


Boxplot Analysis

Control vs Non-Ischemic CM



Control vs Ischemic CM



Biological Interpretation

Several cardiomyopathy-associated genes exhibited differential expression patterns.

Key genes identified included:

- TTN
- ACTN2

- DES
- FLNC
- CRYAB
- FHOD3
- PDLIM3
- PDLIM5
- MYOZ2

These genes are involved in:

- Sarcomere organization
- Cytoskeletal integrity
- Cardiac contractility
- Mechanical signaling
- Heart failure progression

Discussion

Integration of six independent transcriptomic studies substantially improved statistical power and biological consistency.

ComBat correction reduced platform-specific batch effects while Limma identified robust disease-associated transcriptional signatures.

PCA and heatmap analyses demonstrated clear separation between disease and control samples.

Conclusion

This study successfully integrated six GEO microarray datasets to identify transcriptional alterations associated with cardiomyopathy.

Major analytical steps included:

1. GEO Data Collection
2. Gene Annotation
3. Dataset Integration
4. Batch Effect Correction
5. Differential Expression Analysis
6. PCA
7. Heatmap Analysis
8. Volcano Plot Analysis

The identified genes represent potential biomarkers and therapeutic targets for future experimental validation.

Reproducibility

Source scripts used in this project:

- download_data.R
- cardiomyopathy_analysis.R

Session Information

```
## R version 4.4.0 (2024-04-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 10 x64 (build 19045)
##
## Matrix products: default
##
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: Asia/Karachi
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] dplyr_1.1.4   ggplot2_4.0.0 knitr_1.50
##
## loaded via a namespace (and not attached):
## [1] vctrs_0.6.5      cli_3.6.2      rlang_1.1.3     xfun_0.53
## [5] png_0.1-8        generics_0.1.4 S7_0.2.0        glue_1.7.0
## [9] htmltools_0.5.8.1 scales_1.4.0    rmarkdown_2.29  grid_4.4.0
## [13] evaluate_1.0.5   tibble_3.3.0    fastmap_1.2.0   yaml_2.3.10
## [17] lifecycle_1.0.4  compiler_4.4.0 RColorBrewer_1.1-3 pkgconfig_2.0.3
## [21] rstudioapi_0.17.1 farver_2.1.2    digest_0.6.36   R6_2.6.1
## [25] tidyselect_1.2.1 dichromat_2.0-0.1 pillar_1.11.0   magrittr_2.0.3
## [29] jpeg_0.1-10      withr_3.0.2     tools_4.4.0     gtable_0.3.6
```